

Sathwik HS

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Education

PES University

B.Tech, CSE

Bengaluru

2023-27

Relevant Coursework:

100x Devs Cohort 3 ; TUF Strivers A2Z DSA Course ; CWH Complete Python

TLC PU College

Pre-University, PCMS

Mangaluru

2021-23

Experience

PESU ISFCR

Research Intern

PESU, Bengaluru

June 2025 - July 2025

- Engineered and optimized Solidity smart contracts to issue 100% tamper-proof, soulbound NFTs for a secure academic and professional certification platform.
- Architected a decentralized application (dApp) ecosystem utilizing the MERN stack to establish an immutable, cryptographically verifiable audit trail for credentials.
- Automated the credential validation workflow via smart contract execution, reducing verification time from multiple days to under 5 seconds.
- Integrated front-end web interfaces with backend blockchain networks to streamline the end-to-end certificate issuance and verification lifecycle.

Projects

Aethel

Decentralized AI Version Control

August 2025 – Present

- Architected a decentralized version control and audit system to establish an immutable cryptographic trail for AI model fine-tuning
- Engineered a storage-efficient tracking mechanism utilizing Low-Rank Adaptation (LoRA) patches, eliminating base model duplication
- Integrated blockchain infrastructure to manage distributed storage of model weights and strictly prevent unauthorized alterations
- Designed the consensus and storage layer to authenticate and retrieve lightweight AI adapter patches under highly concurrent network conditions

DecentraCert

Decentralized Certificate and Credential Platform

June 2025 – July 2025

- Developed a decentralized application (dApp) enabling issuers to generate forgery-proof blockchain certificates using smart contracts
- Optimized Solidity contracts and implemented batch issuance utilizing Merkle proofs, reducing overall gas costs by 85%
- Integrated MetaMask visibility and automated discovery protocols, allowing Soulbound NFTs to

appear directly in user wallets

- Built a public verification system with one-click cryptographic validation to ensure rapid transparency and trust in credentialing

Terrain Classification for Small Legged Robots

Deep Learning & Robotics

October 2025

- Developed a deep learning pipeline in Python to classify diverse terrain types for robotic systems utilizing raw tactile sensor data
- Engineered neural network architectures to process high-frequency data streams, enabling real-time classification and adaptive mobility
- Optimized data preprocessing workflows to improve feature extraction and model accuracy across distinct physical surfaces

MiniRAFT

Distributed Systems & Consensus

April 2026

- Developed a deep learning pipeline in Python to classify diverse terrain types for robotic systems utilizing raw tactile sensor data
- Engineered neural network architectures to process high-frequency data streams, enabling real-time classification and adaptive mobility
- Optimized data preprocessing workflows to improve feature extraction and model accuracy across distinct physical surfaces

VaidyaDhara

AI-Powered Public Health Platform

September 2025

- Developed a deep learning pipeline in Python to classify diverse terrain types for robotic systems utilizing raw tactile sensor data
- Engineered neural network architectures to process high-frequency data streams, enabling real-time classification and adaptive mobility
- Optimized data preprocessing workflows to improve feature extraction and model accuracy across distinct physical surfaces

Leadership & Activities

F_SOCI3TY - CTF Team

Bengaluru

Co-Founder & Member

November 2025 – Present

- Secured Runner-up position at the IIT Hyderabad Capture The Flag (CTF) competition in January 2026 by successfully exploiting complex vulnerabilities
- Participated in 10+ distinct, high-difficulty national and international CTFs across cryptography, steganography, blockchain, OSINT and web exploitation categories

Skills & Interests

Technical: Python, JavaScript, MERN, Solidity

Language: Kannada, English, Hindi

Interests: CTFs, Hackathons